Microwaved nickel silicon carbide catalysts for the methane dry reforming reaction

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Abstract

Methane (CH_4) and carbon dioxide (CO_2) are two greenhouse gases (GHG) that can be combined during the CH_4 dry reforming (DRM) reaction to produce hydrogen (H_2) and carbon monoxide (CO), a mixture of gases of high energetic value. Traditional catalysts use for this reaction such as nickel supported on alumina, zeolites or silica catalysts have been reported to favour carbon deposition and nickel (Ni) sintering because of their low thermal conductivity. Beta silicon carbide (β-SiC) catalysts could represent an opportunity for the DRM reaction because of their interesting properties for high temperature chemical processes, among them its high thermal conductivity and its microwave absorbance property. The later could be used to affect positively Ni-Silicon carbide based catalysts for DRM.

In this work, microwave irradiation (MIR) was applied as a post-treatment method for Ni- β-SiC and Ni-Ti- Cβ-SiC catalysts to obtain more active, stable, and selective catalysts. In the case of Ni- β-SiC catalysts, the MIR strengthened the Ni and β-SiC interaction at high temperatures preventing Ni particles from sintering while for the Ni-Ti- Cβ-SiC catalysts, the Ni and Ti-Cβ-SiC interactions were weakened. From the catalytic testing, firstly it was observed that, the Ni-Ti- Cβ-SiC catalyst showed better performance than the Ni- β-SiC catalyst due to the higher surface area of the Ti- Cβ-SiC support over the β-SiC. Secondly, for the β-SiC support, the MIR in some cases (10 s and 25 s exposure) improved significantly the CH_4, CO_2 conversions, and the H_2/CO ratios as well as the catalyst's stability.

On the other hand, the untreated Ni-Ti- Cβ-SiC catalyst showed better performance than the Ni-Ti- Cβ-SiC microwaved catalysts but displayed a quicker deactivation behaviour at 15 h on stream, while all the Ni- Ti- Cβ-SiC microwaved ones were stable during the DRM.

In both cases of Ni- β-SiC, and Ni-Ti- Cβ-SiC catalysts, of 30 seconds exposure to MIR was detrimental to the performance of the catalysts for the DRM process, which is an indication that there is a MIR exposure time that must not be exceeded for the catalyst post-treatment to be beneficial.

For future work, the effect of iron (Fe) as a chemical promoter for Ni silicon carbide catalysts will be investigated since Fe is also a good microwave absorber compound that can alter the catalytic behaviour of Ni catalysts.

Key words: Carbon dioxide, methane, dry reforming, Microwave irradiation, Ni Silicon carbide catalysts